

Github url: https://github.com/asnashameel/Devops_main_Project.git

S3 bucket url: https://365591124323-terraform-state-7736.s3.us-west-2.amazonaws.com/cloudops/Devops_main_Project-main.zip

Cloud-Ops in Action: Deploying a Multi-Tier User Portal

INTRODUCTION:-

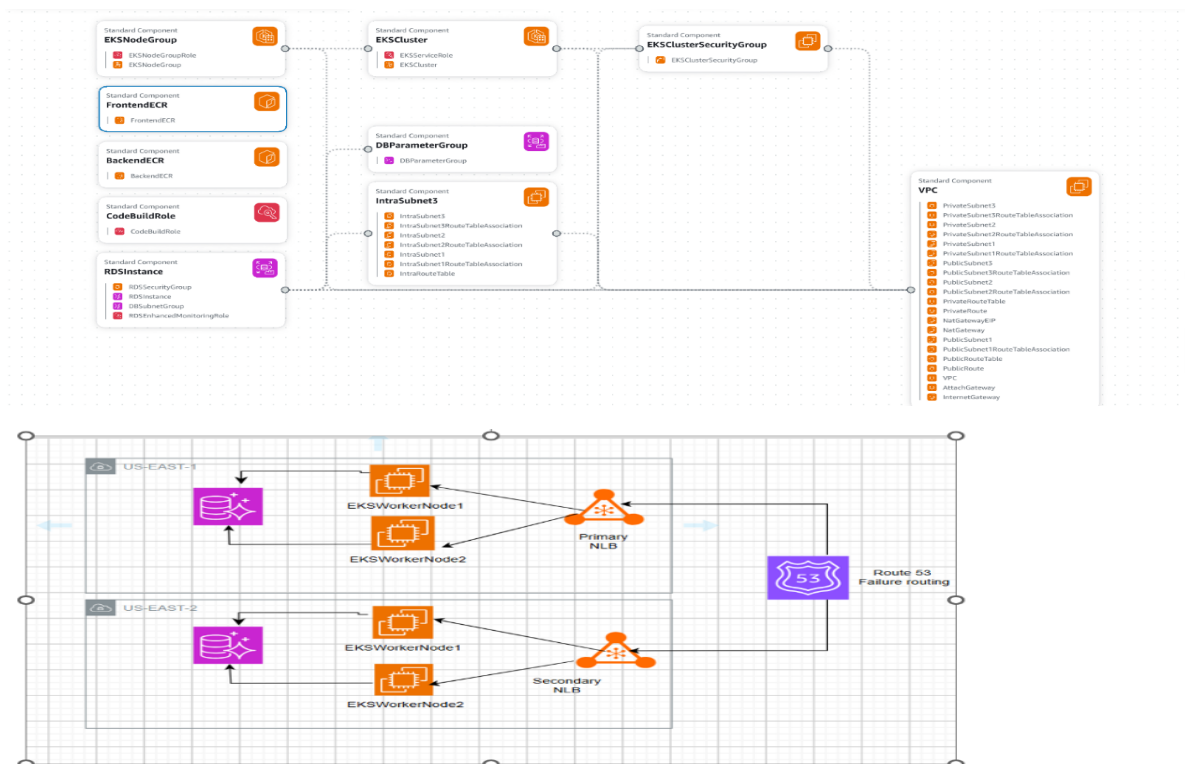
This document provides a comprehensive overview of the **cloudops-demo** three-tier application. It details the architecture, technologies used in each layer, deployment strategy on AWS, and key components, along with guidance for understanding and troubleshooting the system.

Project Purpose:- A web application for managing user profiles, responsive frontend and robust backend.

Three tier Architecture Overview :-

The application is deployed on AWS, leveraging Amazon Elastic Kubernetes Service (EKS) for container orchestration, Amazon RDS for the relational database, and various networking and IAM components to ensure secure and scalable operations.

- **Frontend:** React application
- **Backend:** Node.js/Express API
- **Database:** AWS RDS PostgreSQL



Phase 1:-Deploy an multi tier Application on EKS Cluster

Initially created git repo Devops_main_Project

```

root@ip-172-31-95-241:/home/ubuntu# git clone git@github.com:asnashameel/Devops_main_Project.git
Cloning into 'Devops_main_Project'...
remote: Enumerating Objects: 47, done.
remote: Counting objects: 100% (47/47), done.
remote: Compressing objects: 100% (41/41), done.
Receiving objects: 100% (47/47), 25.49 KiB | 12.74 MiB/s, done.
Resolving deltas: 100% (1/1), done.
remote: Total 47 (delta 1), reused 47 (delta 1), pack-reused 0 (from 0)
root@ip-172-31-95-241:/home/ubuntu# ls
3-tier-dep Devops_main_Project awscli2.zip get-docker.sh
root@ip-172-31-95-241:/home/ubuntu# cd Devops_main_Project/
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project# ls
README.md backend buildspec.yml database docker-compose.yml env.example frontend infrastructure k8s scripts terraform
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project# cd frontend
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/frontend# ls
Dockerfile nginx.conf package.json public src
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/frontend# nano Dockerfile
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/frontend#

```

```

root@ip-172-31-95-241:/home/ubuntu# git clone git@github.com:asnashameel/Devops_main_Project.git
Cloning into 'Devops_main_Project'...
remote: Enumerating Objects: 47, done.
remote: Counting objects: 100% (47/47), done.
remote: Compressing objects: 100% (41/41), done.
Receiving objects: 100% (47/47), 25.49 KiB | 12.74 MiB/s, done.
Resolving deltas: 100% (1/1), done.
remote: Total 47 (delta 1), reused 47 (delta 1), pack-reused 0 (from 0)
root@ip-172-31-95-241:/home/ubuntu# ls
3-tier-dep Devops_main_Project awscli2.zip get-docker.sh
root@ip-172-31-95-241:/home/ubuntu# cd Devops_main_Project/
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project# ls
README.md backend buildspec.yml database docker-compose.yml env.example frontend infrastructure k8s scripts terraform
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project# cd frontend
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/frontend# ls
Dockerfile nginx.conf package.json public src
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/frontend# nano Dockerfile
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/frontend#

```

Created docker image for my frontend application by using docker build -t frontend .

```

root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/frontend# docker build -t dev-frontend .
[+] Building 175.0s (16/16) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 683B
=> [internal] load metadata for docker.io/library/nginx:alpine
=> [internal] load metadata for docker.io/library/node:18-alpine
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [builder 1/6] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e
=> [stage-1 1/4] FROM docker.io/library/nginx:alpine@sha256:65645c7bb6a0661892a8b03b89d0743208a18dd2f3f17a54ef4b76fb8e2f2a10
=> [internal] load build context
=> => transferring context: 893B
=> CACHED [stage-1 2/4] COPY nginx.conf /etc/nginx/conf.d/default.conf
=> CACHED [builder 2/6] WORKDIR /app
=> CACHED [builder 3/6] COPY package*.json ./
=> [builder 4/6] RUN npm install
=> [builder 5/6] COPY . .
=> [builder 6/6] RUN npm run build
=> [stage-1 3/4] COPY --from=builder /app/build /usr/share/nginx/html
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/frontend/public# docker ps -a

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
49bd34737f2c	207567798584.dkr.ecr.us-east-1.amazonaws.com/dev-frontend	"/docker-entrypoint..."	20 seconds ago	Up 20 seconds
bacc8afde4b7	207567798584.dkr.ecr.us-east-1.amazonaws.com/dev-frontend	admiring_brattain	11 minutes ago	Exited (1) 11 minutes ago
cd24b00dfa61	207567798584.dkr.ecr.us-east-1.amazonaws.com/3-tier-frontend:latest	vibrant_ganguly	3 days ago	Exited (0) 27 hours ago

```

root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/frontend/public# cd ..

```

Tag the image

```

root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/frontend# docker tag dev-frontend 207567798584.dkr.ecr.us-east-1.amazonaws.com/dev-frontend:latest
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/frontend# docker push 207567798584.dkr.ecr.us-east-1.amazonaws.com/dev-frontend:latest
The push refers to repository [207567798584.dkr.ecr.us-east-1.amazonaws.com/dev-frontend]
5a9884b19b06: Pushed
69a9ffc140c3: Pushed
03fcfef73e03: Pushed
0d853d50b128: Pushed
947e805a4ac7: Pushed
811a4dbbf4a5: Pushed
b8d7d1d22634: Pushed
e244aa659f61: Pushed
c56f134d3805: Pushed
d71eae0084c1: Pushed
08000c18d16d: Pushed
latest: digest: sha256:51dce67f288098bdae3bd2f414bac5c69a3ad51ff94a5a8449483136048e7c size: 2615
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/frontend#
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/frontend# docker tag dev-frontend 207567798584.dkr.ecr.us-east-1.amazonaws.com/dev-frontend:latest
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/frontend# docker push 207567798584.dkr.ecr.us-east-1.amazonaws.com/dev-frontend:latest
The push refers to repository [207567798584.dkr.ecr.us-east-1.amazonaws.com/dev-frontend]
5a9884b19b06: Pushed
69a9ffc140c3: Pushed
03fcfef73e03: Pushed
0d853d50b128: Pushed
947e805a4ac7: Pushed
811a4dbbf4a5: Pushed
b8d7d1d22634: Pushed
e244aa659f61: Pushed
c56f134d3805: Pushed
d71eae0084c1: Pushed
08000c18d16d: Pushed
latest: digest: sha256:51dce67f288098bdae3bd2f414bac5c69a3ad51ff94a5a8449483136048e7c size: 2615
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/frontend#

```

```

root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/Frontend# docker build -t dev-frontend .
[+] Building 175.0s (16/16) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.1s
=> => transferring dockerfile: 683B                                0.1s
=> [internal] load metadata for docker.io/library/nginx:alpine    0.1s
=> [internal] load metadata for docker.io/library/node:18-alpine  0.1s
=> [internal] load .dockerignore                                   0.1s
=> => transferring context: 2B                                     0.1s
=> [builder 1/6] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e 0.1s
=> [stage-1 1/4] FROM docker.io/library/nginx:alpine@sha256:65645c7bb6a0661892a8b03b89d0743208a18dd2f3f17a54ef4b76fb8e2f2a10 0.1s
=> [internal] load build context                                   0.1s
=> => transferring context: 893B                                    0.1s
=> CACHED [stage-1 2/4] COPY nginx.conf /etc/nginx/conf.d/default.conf 0.1s
=> CACHED [builder 2/6] WORKDIR /app                               0.1s
=> CACHED [builder 3/6] COPY package*.json ./                      0.1s
=> [builder 4/6] RUN npm install                                   99.1s
=> [builder 5/6] COPY . .                                          0.1s
=> [builder 6/6] RUN npm run build                                72.1s
=> [stage-1 3/4] COPY --from=builder /app/build /usr/share/nginx/html 0.1s
=> [stage-1 4/4] RUN sleep 2000                                    0.1s
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/Frontend/public# docker run -d -p 3000:3000 207567798584.dkr.ecr.us-east-1.amazonaws.com/dev-frontend sleep 2000
49bd347372ce7963adb86dc125f838478557e731851c4d16dfefac188d4e9c6

```

container created by using image

```

[+] Building 13.3s (13/13) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 584B                                0.0s
=> [internal] load metadata for docker.io/library/node:18-alpine  0.1s
=> [internal] load .dockerignore                                   0.0s
=> => transferring context: 2B                                     0.0s
=> [1/8] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e 0.0s
=> [internal] load build context                                   0.0s
=> => transferring context: 7.33kB                                 0.0s
=> CACHED [2/8] WORKDIR /app                                       0.0s
=> [3/8] RUN addgroup -g 1001 -S nodejs                            0.2s
=> [4/8] RUN adduser -S nodejs -u 1001                             0.2s
=> [5/8] COPY package*.json ./                                     0.1s
=> [6/8] RUN npm install                                           7.7s
=> [7/8] COPY . .                                                  0.0s
=> [8/8] RUN chown -R nodejs:nodejs /app                          4.3s
=> exporting image                                                0.5s
=> => exporting layers                                           0.4s
=> => writing image sha256:fa11f7ae12c99bb3d284cd0d2f317cfe4bd1c3cd39b9238deb49fdb914d93ea 0.0s
=> => naming to docker.io/library/dev-backend                    0.0s
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/backend#

```

Build docker image for frontend,tagged the image and pushed to ecr repo

```

root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/backend# docker tag dev-backend 207567798584.dkr.ecr.us-east-1.amazonaws.com/dev-backend:latest
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/backend# docker push 207567798584.dkr.ecr.us-east-1.amazonaws.com/dev-backend:latest
The push refers to repository [207567798584.dkr.ecr.us-east-1.amazonaws.com/dev-backend]
af32f1b7d4a7: Pushed
545c6ea37ffa: Pushed
123c077b176d: Pushed
5513b7967802: Pushed
f53fc8f26ed6: Pushed
ca5cd0fd36fa: Pushed
777dc634dbf5: Pushed
82140d9a70a7: Pushed
f3b40b0cdb1c: Pushed
0b1f26057bd0: Pushing [=====>] 74.58MB/113.9MB
08000c18d16d: Pushed

```

Next are the deployment steps

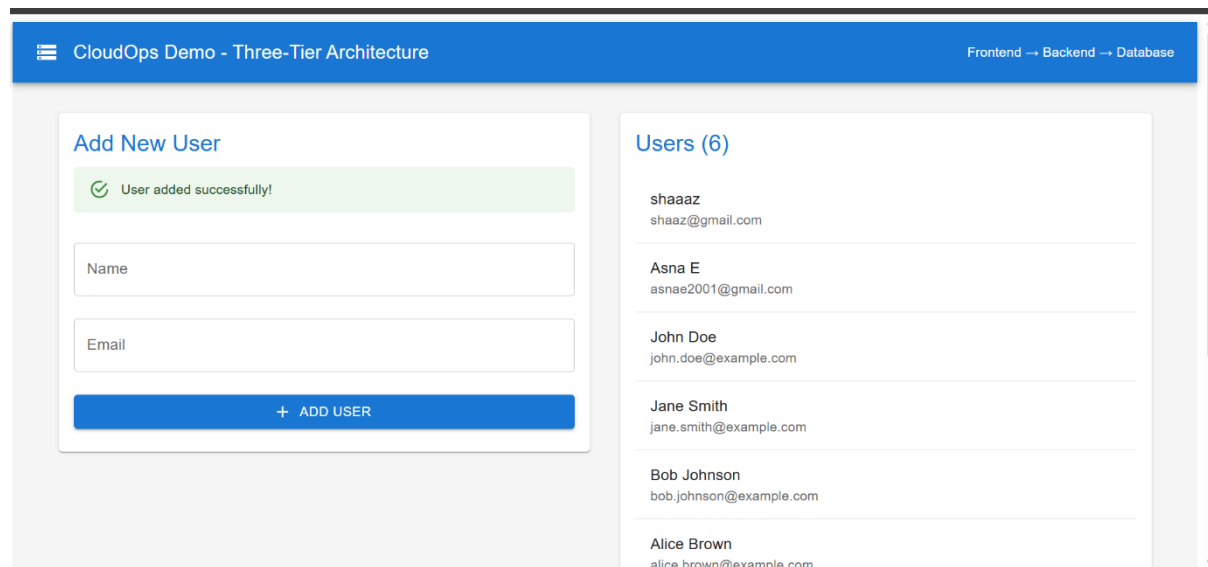
Initially created cluster

Deployed all yaml files

```

root@ip-172-31-95-241:/home/ubuntu# eksctl create cluster --name kube-cluster --region us-east-1 --node-type t2.medium --nodes-min 2 --nodes-max 2
2025-06-19 16:51:09 [i] eksctl version 0.210.0
2025-06-19 16:51:09 [i] using region us-east-1
2025-06-19 16:51:09 [i] setting availability zones to [us-east-1c us-east-1d]
2025-06-19 16:51:09 [i] subnets for us-east-1c - public:192.168.0.0/19 private:192.168.64.0/19
2025-06-19 16:51:09 [i] subnets for us-east-1d - public:192.168.32.0/19 private:192.168.96.0/19
2025-06-19 16:51:09 [i] nodegroup "ng-6659c31a" will use "" [AmazonLinux2023/1.32]
2025-06-19 16:51:09 [i] using Kubernetes version 1.32
2025-06-19 16:51:09 [i] creating EKS cluster "kube-cluster" in "us-east-1" region with managed nodes
2025-06-19 16:51:09 [i] will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2025-06-19 16:51:09 [i] if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=us-east-1 --cluster=kube-cluster'
2025-06-19 16:51:09 [i] Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "kube-cluster" in "us-east-1"
2025-06-19 16:51:09 [i] CloudWatch logging will not be enabled for cluster "kube-cluster" in "us-east-1"
2025-06-19 16:51:09 [i] you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --region=us-east-1 --cluster=kube-cluster'
2025-06-19 16:51:09 [i] default add-ons vpc-cni, kube-proxy, coredns, metrics-server were not specified, will install them as EKS add-ons
2025-06-19 17:04:54 [i] cluster should be functional despite missing (or misconfigured) Client Binaries
2025-06-19 17:04:54 [✓] EKS cluster "kube-cluster" in "us-east-1" region is ready
root@ip-172-31-95-241:/home/ubuntu# aws eks update-kubeconfig --region us-east-1 --name kube-cluster
Added new context arn:aws:eks:us-east-1:207567798584:cluster/kube-cluster to /root/.kube/config
root@ip-172-31-95-241:/home/ubuntu# kubectl get nodes
NAME                                STATUS    ROLES    AGE   VERSION
ip-192-168-24-77.ec2.internal        Ready    <none>    16m   v1.32.3-eks-473151a
ip-192-168-55-63.ec2.internal        Ready    <none>    16m   v1.32.3-eks-473151a
root@ip-172-31-95-241:/home/ubuntu# cd Devops_main_Project/
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project# ls
README.md  backend  buildspec.yml  database  docker-compose.yml  env.example  frontend  infrastructure  k8s  scripts  terraform
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project# cd k8s/
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/k8s# ls
00-namespaces.yaml  02-secrets.yaml  08-backend-service.yaml  10-frontend-service.yaml  12-hpa.yaml
01-configmap.yaml  07-backend-deployment.yaml  09-frontend-deployment.yaml  11-ingress.yaml  13-network-policy.yaml
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/k8s# nano 09-frontend-deployment.yaml
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/k8s# kubectl create namespace cloudops-demo
namespace/cloudops-demo created
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/k8s#

```



Accessed the application by using host address

Found application is working fine

Phase2:- Automate the Previous deployment using CodePipeline

1) used aws code commit as source in pipeline to trigger the pushes

Initially created one repository for aws code commit

Repositories [Info](#)

Notify

Clone URL

View repository

Delete repository

Create repository

1

	Name	Description	Last modified	Clone URL	AWS KMS Key
☐	cloudops-demo	-	4 hours ago	HTTPS SSH HTTPS (GRC)	arn:aws:kms:us-east-1:365591124323:key/ee874354-d669-4782-9212-68ad7b425130

Cloned the repository

```
gduud9JCpfX59q3pnfULB4IdAKOBbrCh1MZQZXFd/4gZ2oW7uvXLReWWjy3AAVjzzWr39m2dLLYp7vpl2bAxqCDstHXJ8K2S8x/f81b//BFGYJpBf1xWKO7g9NZg8V/ewToKq31SgcFZ
PeYznmTBLgsVDG6hhTJksrjotWtGfh8fVXxaEFckdIKM7IohAOZUVYP1bILrUvsn21IdLWFxPU6u+FmMYFxfjL0hycV93721a5s000Zv/bMyMoNywQ007/PMkOTRmqp/8prAQSPb/PTY
nSpjIoLvabryunk88VuK4gay7hwpB7zeY70yx5tgf7oKgVpx7HGFNVu2waVPfwc8U3lxRM/79SASQGWtLLFsr+5d2AJ0K78qWwCV6YUfG6AveNX2d8zbYguzjItvYyPEm2MRy1Z0g36X
lZf425gjZ7pDx7mXchtQEg== codecommit-access
root@ip-172-31-86-56:~/.ssh# cd
root@ip-172-31-86-56:~# cd /home/ubuntu/
root@ip-172-31-86-56:/home/ubuntu# nano ~/.ssh/config
root@ip-172-31-86-56:/home/ubuntu# chmod 600 ~/.ssh/config
root@ip-172-31-86-56:/home/ubuntu# git clone ssh://git-codecommit.us-east-1.amazonaws.com/v1/repos/cloudops-demo
Cloning into 'cloudops-demo'...
The authenticity of host 'git-codecommit.us-east-1.amazonaws.com (52.94.226.180)' can't be established.
RSA key fingerprint is SHA256:eLMY1j0DKA4uvDZcl/KgtIayZANwX6t8+8isPtotBoY.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'git-codecommit.us-east-1.amazonaws.com' (RSA) to the list of known hosts.
warning: You appear to have cloned an empty repository.
root@ip-172-31-86-56:/home/ubuntu# ssh git-codecommit.us-east-1.amazonaws.com
You have successfully authenticated over SSH. You can use Git to interact with AWS CodeCommit. Interactive shells are not supported.Connection
on to git-codecommit.us-east-1.amazonaws.com closed by remote host.
Connection to git-codecommit.us-east-1.amazonaws.com closed.
root@ip-172-31-86-56:/home/ubuntu#
```

Uploaded ssh key to repository

SSH public keys for AWS CodeCommit (1)

Actions

Upload SSH public key

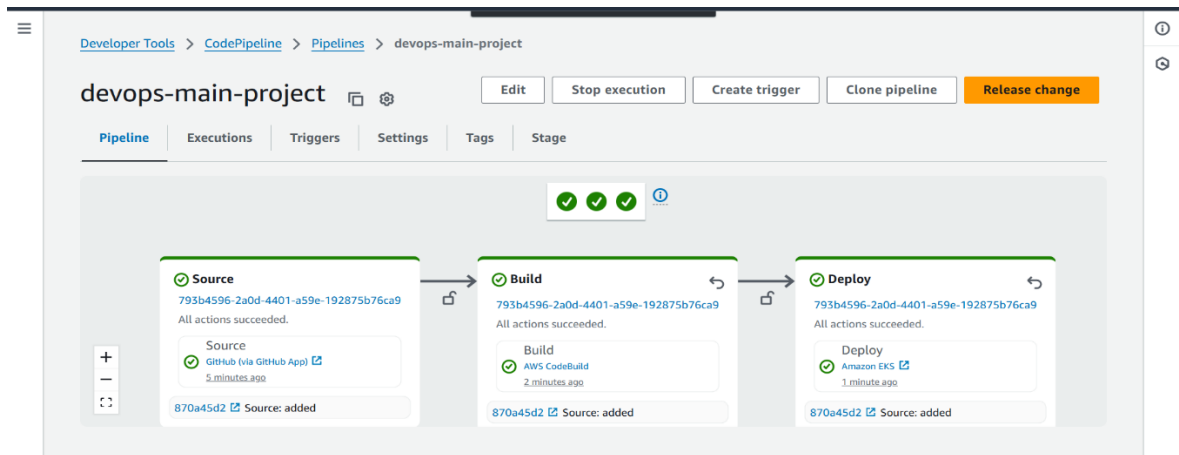
Use SSH public keys to authenticate access to AWS CodeCommit repositories. You can have a maximum of five SSH public keys (active or inactive) at a time. [Learn more](#)

	SSH Key ID	Uploaded	Status
☐	APKAVKHxN3FR673XTN4N	Now	Active

Then push will automatically trigger

Added buildspec.yml to build stage

```
phases:
  install: # Install kubectl, Helm, Docker
  pre_build: # Run tests, login to ECR
  build: # Build Docker images, tag with commit hash
  post_build: # Push to ECR, deploy to AWS
```

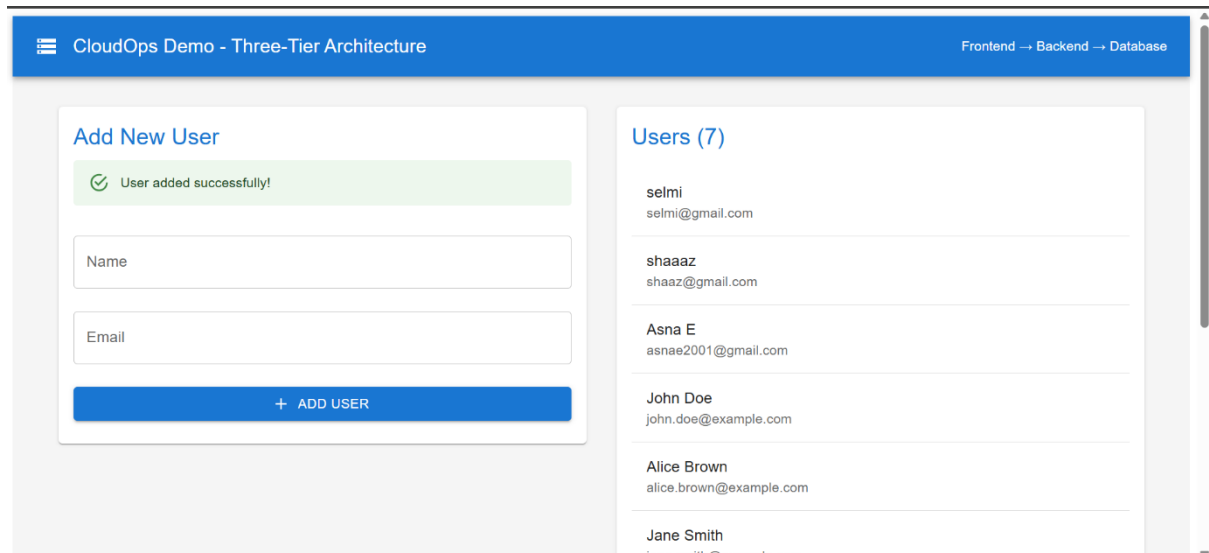


Kubernetes manifests files are added in manifests in, It will deploy all yaml files in kubernetes manifests

```
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/k8s# kubectl get ing -n cloudops-d
NAME          CLASS    HOSTS
cloudops-ingress <none>  a4160f7605eda49cfa12fbfbbcd5a3a6-1839d92ed829f63a.elb.us-east-1.amazonaws.com
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/k8s#
```

Kubectrl get ing -n cloudops-demo,will get host address of ingress

```
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/k8s# kubectl get ing -n cloudops-demo
NAME          CLASS    HOSTS
cloudops-ingress <none>  a4160f7605eda49cfa12fbfbbcd5a3a6-1839d92ed829f63a.elb.us-east-1.amazonaws.com
root@ip-172-31-95-241:/home/ubuntu/Devops_main_Project/k8s#
```

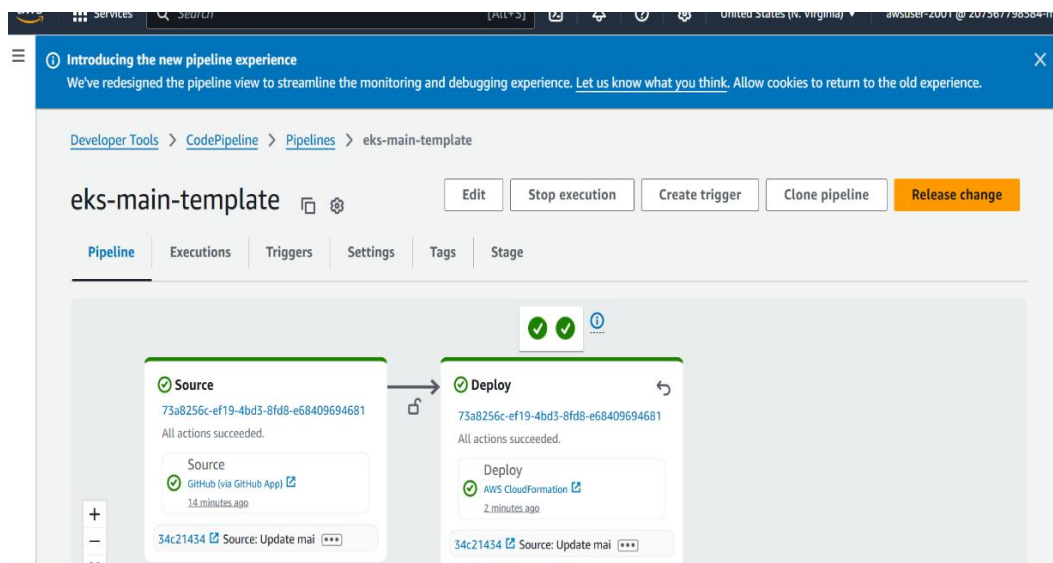
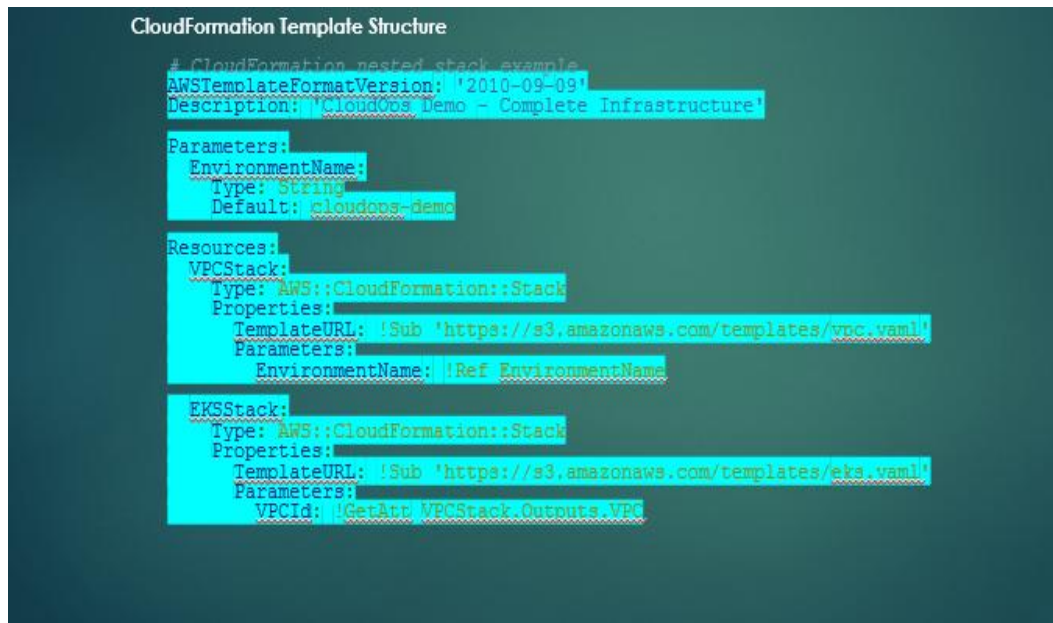


Accessed the application, it is working fine

Phase3:-Expose the deployment in multiple AWS region (Terraform in one region and cloudformation in another region)

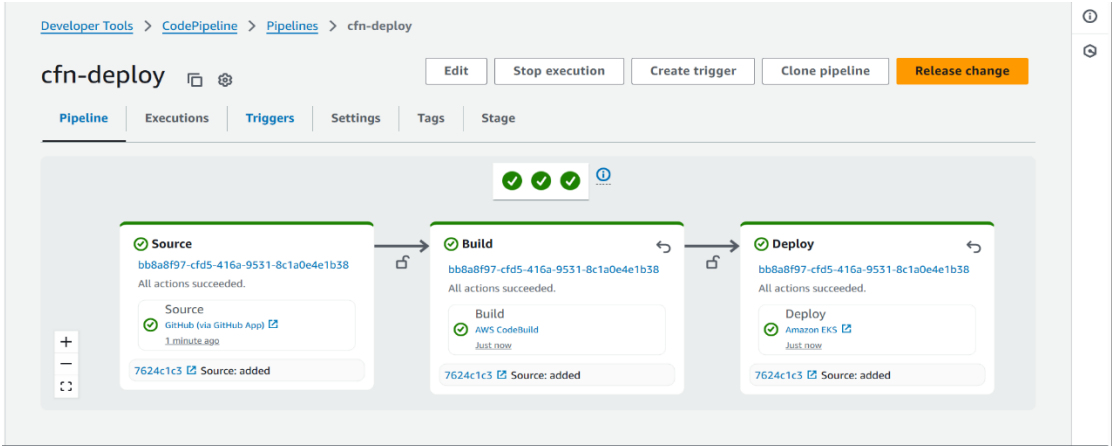
Created infrastructure using cloudformation in us-east-1 region

Created template for that eks-main-template.yaml



Initially created one pipeline for creating infrastructure and deployed in AWS cloudformation

Then build and deploy the project in eks using another pipeline



Below is the image showing stack created using cloudformation

Filter by stack name

Active

☒ View nested

< 1 >

	Stack name	Status	Created time	Description
☐	eksctl-cloudops-demo-addon-iamserviceaccount-amazon-cloudwatch-cloudwatch-agent	CREATE_COMPLETE	2025-06-24 10:37:21 UTC+0530	IAM role for ser "amazon-cloudwatch/clo [created and m
☐	eksctl-dotnet-addon-iamserviceaccount-kube-system-aws-load-balancer-controller	CREATE_COMPLETE	2025-06-23 10:05:00 UTC+0530	IAM role for ser "kube-system/a balancer-contrc and managed b
☐	asna-cloud-stack	CREATE_COMPLETE	2025-06-23 10:01:52 UTC+0530	Complete Clou infrastructure w ECR, and suppo
				EKS Cluster with

It will automatically create ecr, eks,vpc subnets and all infrastructure

ECR

zon Elastic

ainer Registry

te registry

itories

res & Settings

c registry

itories

gs

Private repositories (4)

View push commands

Delete

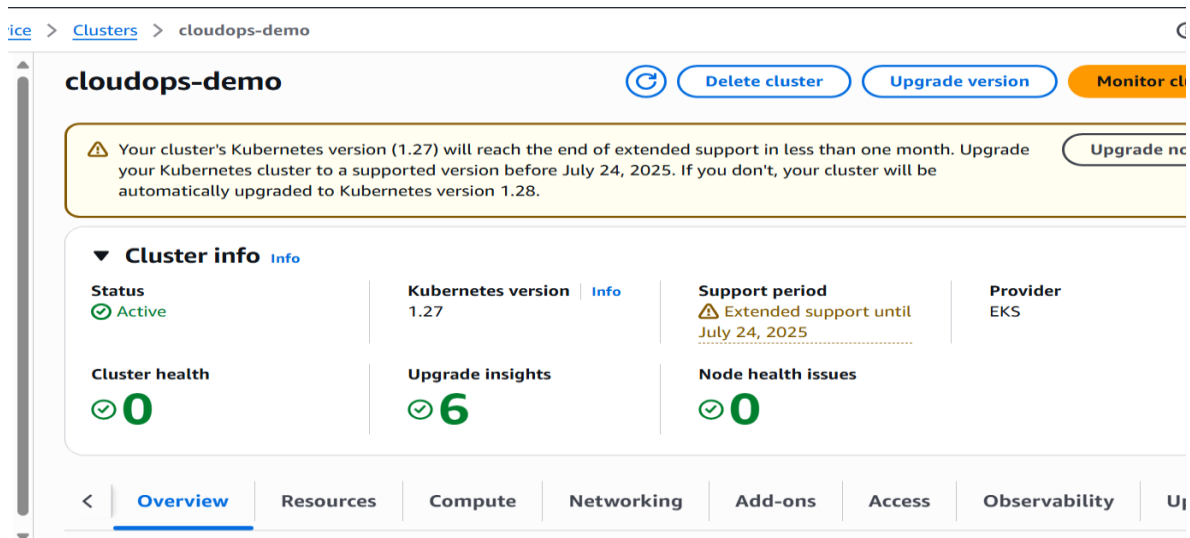
Actions ▾

Create repository

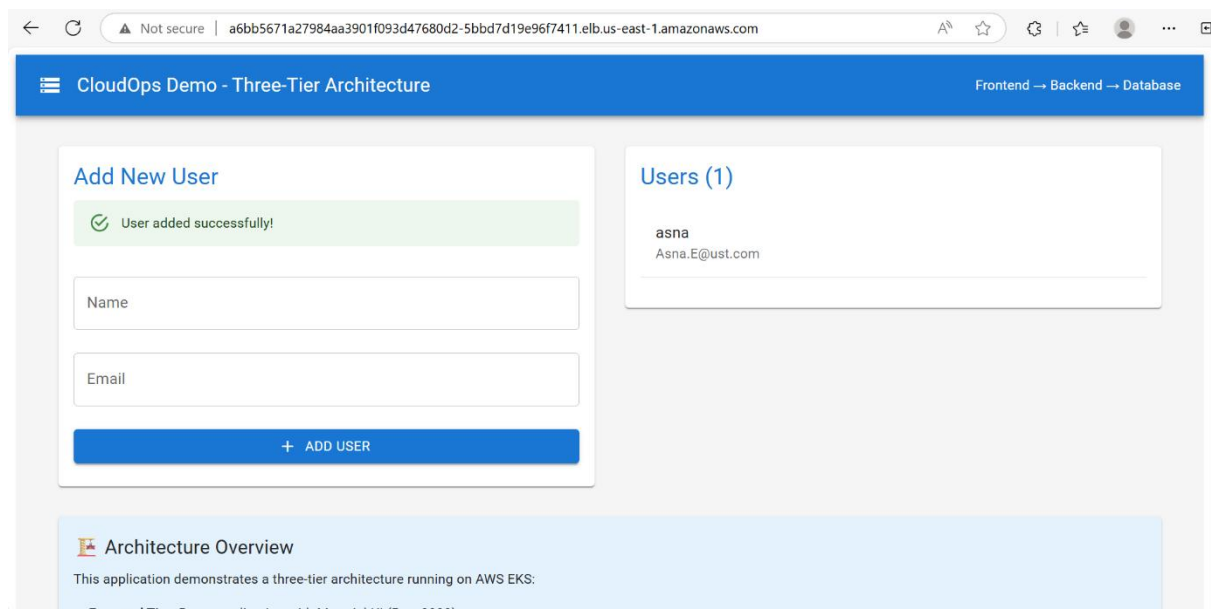
Q Search by repository substring

	Repository name ▲	URI	Created at ▾	Tag immutability	Encryption type
☐	cloudops-backend	 715841362372.dkr.ecr.us-east-1.amazonaws.com/cloudops-backend	June 23, 2025, 10:01:56 (UTC+05.5)	Mutable	AES-256
☐	cloudops-frontend	 715841362372.dkr.ecr.us-east-1.amazonaws.com/cloudops-frontend	June 23, 2025, 10:01:56 (UTC+05.5)	Mutable	AES-256

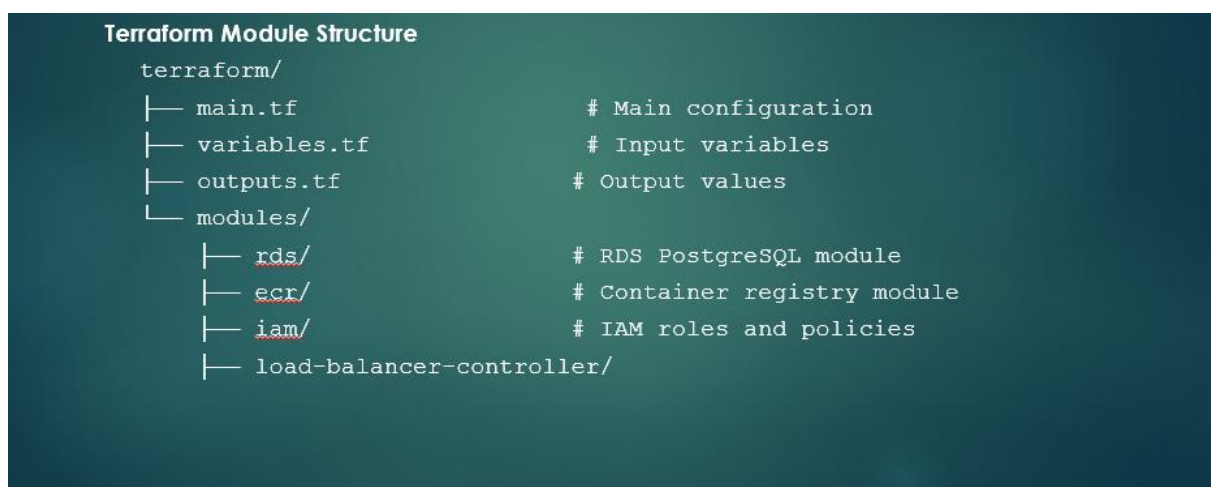
EKS



Accessed the application , yeah it is working fine



Terraform:-



Created terraform infrastructure using one pipeline that contain buildspec.yaml for terraform init plan and apply

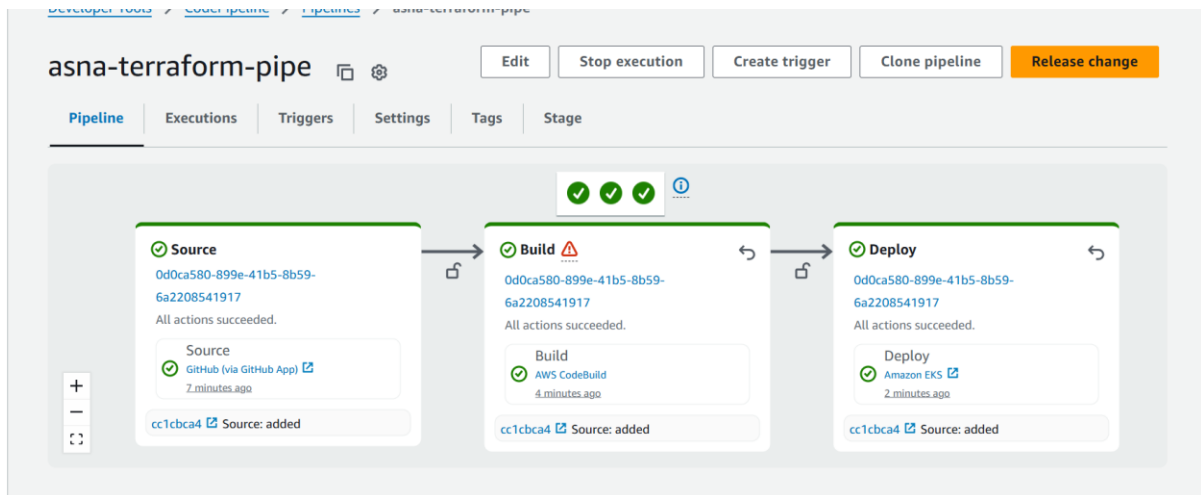
```

fi

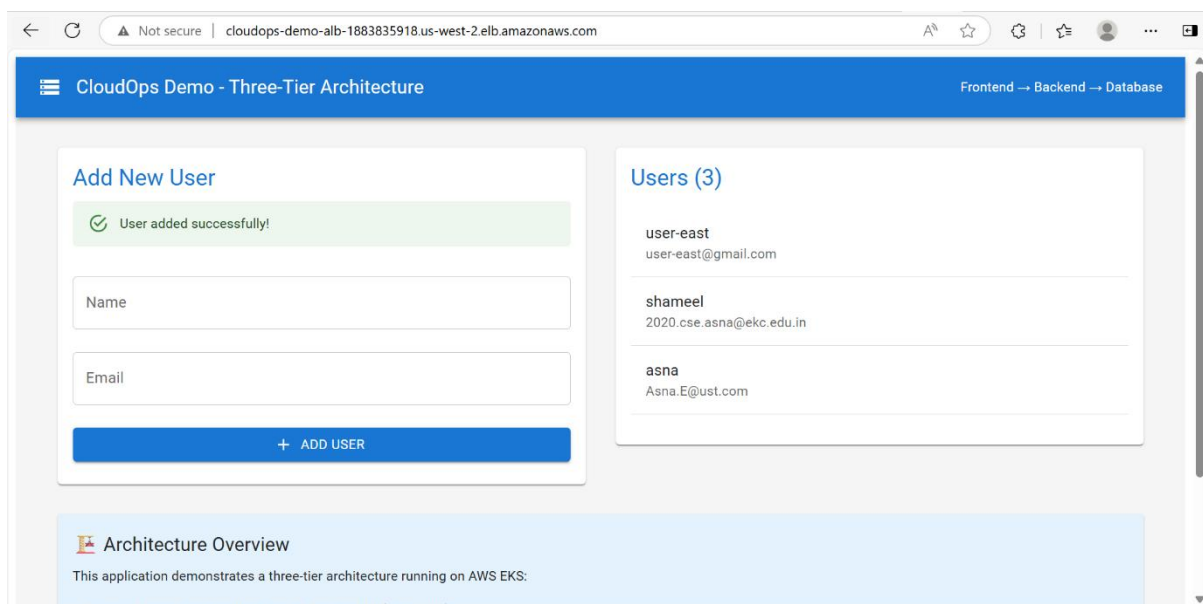
=== Infrastructure Outputs ===
cluster_certificate_authority_data =
  "LS0tLS1CRUdJc1IbDRVJUSzYzQ0RURSc0tLS0tCk15SURCVENDQWUyZ0F3SUJBZ0ZlJT1t1HZwtYN1FMYWt3RFFZSKstvWk1odmNOQVFFTEJRQXGdG
VEVUTUJ3FR0EzVUUXQXhNS2EzZVml1aWEP1wl1h5BgNM6QWgdzB5T1RBmK1q5XhNek13tRKYUz3MHP0VEEYtWpBeE16STFOREphTUjVeApFekFSQz
mdOVKcJBTVRDbzQxXW1lWwZjVjBwE15Zp1d4Ua1BMEddDU3FUHU0tIMRRRUJBUVVBQTRJQkR3QXdnZ0VLCkFvSUJBUURZOHDneUVZcnBEdmpodX
xwGdR0W5OZGq5GdLdRXL1MakRkXQ1K1Sxm9OOVPWtJFTdFHMm13eH1bk4KMDZGdFsktFNEF3djYvU29mV1ZKZ0vOXEWz24wRE9jMU5nU011L3p
mST1NZU1ZwMhtHk1tM1EpLaEraaHfSZ0ZJOKwPQc2F5Mi51UXg502xESFJtZGvqd0RON2Fi1MDg4NmU4ZG61NzhFanJNeFpVNE1Yz1P42NSmT50x1
TURNk9tC29XckRES1VNERMMN12SHRHrTss3Q35bzbJ6cG3y3jBVRG000EZ2Uj3d3XoYsmN3RDgzMFUxYmg3TfWfVialBrMjcKdUNNVHZQ0
TJuNd1Nv3VTWS9Za1o1M1dTRDB6a0jV33RTam9pbnhxWdhDqdkxTeTh2czdoVzZvMXd2SFdsd3NnVQo4T0lhzYtIQVdTzk10dHc0tK84aEJnSk
JDOgKwQmdNmKfBR2pXVEJYtUE0R0EXVWREd0VLC3dRRUF35UNuWERFQCKJnt1ZiUK1CQWY4RUJ3QURBUUGvTUiwR0EXVWREZ1FXQk3sdTZYt2t
22TBuW9oUmQrYz34Q0dGFSFKZQ6qQVYKQmdOVkhSRUVEakFNZ2dwcmaRXSmxjbTVsZedWek1BMEDDU3FUHU0tIMRRRUJDD1VBQTRJQkFRFRBc
ZVF35jFJZApHedDvUTZ3RVJ3bnRRd2ErVDJNQX3J02OHUSZ0pazhHZZVLTHVJc0v1c3BFVTROQ31GYWV1NHbNmWpsKd0RaNuFVcmtn50ZVMXfVt
136HMFdm1qcUmlUjlyV2NUU1JGcU1Z2ejykBENwOEu1aDhLXFPWGFtNGU4R1QJ1RDFvWuXgUdEKcDrtRmj3qmdN5KsNVmW3K20a25mT1JHVh
pLzJdMtdBqL1M3VzE4NFhXVUZHChNnQVUvcz1PaWpVd1Nnd1MxYwpsQUwra3VjbFpxRG5CNmpLKzFnREVXhmdTcmUzYXY2K240V14CYURDaTy
ySTNPVUE2UddwNmhdMStGwNRPdTEi0CLvaZDFBVVRZU1KdHnVeHJ3Sh2hNOgkIN3Z0P0EJOuWZnV1RabmdzCfYzOedjYSEhCME96bDVJ3UVAwOHhF
N1NMUXAQ0dIU2U5OW06172Tci0tLS0tRU5UEWI1JRk1DQVRFLS0t"
cluster_endpoint = "https://5BB8C62322E206B71C17C0EA7E0CC9F1.y14.us-west-2.eks.amazonaws.com"
cluster_iam_role_name = "cloudops-demo-cluster-20250622132054187300000001"
cluster_name = "cloudops-demo"
cluster_oidc_issuer_url = "https://oidc.eks.us-west-2.amazonaws.com/id/5BB8C62322E206B71C17C0EA7E0CC9F1"
cluster_security_group_id = "sg-0c839e7a2573f26e9"
codebuild_role_arn = "arn:aws:iam::715841362372:role/cloudops-demo-codebuild-role"
ecr_repositories = {
  "cloudops-backend" = "715841362372.dkr.ecr.us-west-2.amazonaws.com/cloudops-backend"
  "cloudops-frontend" = "715841362372.dkr.ecr.us-west-2.amazonaws.com/cloudops-frontend"
}
kubect1_config_command = "aws eks update-kubeconfig --region us-west-2 --name cloudops-demo"

```

After creating the infrastructure deployed the application in eks using another pipeline



After successful deployment ,accessed the application,showed it is working fine



Phase4 :- Route53

Route53 failover Architecture



aws [Search] [Alt+S] Global Asna @ 715841362372-mmml

Route 53 > Hosted zones > cloudops-demo-asnae.com

Route 53

- Dashboard
- Hosted zones
- Health checks
- Profiles [New](#)
- ▼ IP-based routing
 - CIDR collections
- ▼ Traffic flow
 - Traffic policies
 - Policy records
- ▼ Domains
 - Registered domains
 - Requests
- ▼ Resolver
 - VPCs

Records (1/2) Info

[Delete record](#) [Import zone file](#) [Create record](#)

The following table lists the existing records in cloudops-demo-asnae.com. You can't delete the SOA record or the NS record named cloudops-demo-asnae.com.

Filter records by p Type Routing p... Alias

	Record ...	Type	Routin...	Differ...
☐	cloudops-...	NS	Simple	-
☒	cloudops-...	SOA	Simple	-

Edit record

Record name
cloudops-demo-asnae.com

Record type
SOA

Value
ns-893.awsdns-47.net. awsdns-hostmaster.amazon.com. 1 7200 900 1209600 86400

Alias
No

TTL (seconds)
900

Routing policy
Simple

aws [Search] [Alt+S] Global Asna @ 715841362372-mmml

Route 53 > Hosted zones

Route 53

- Dashboard
- Hosted zones
- Health checks
- Profiles [New](#)
- ▼ IP-based routing
 - CIDR collections
- ▼ Traffic flow
 - Traffic policies
 - Policy records
- ▼ Domains
 - Registered domains
 - Requests
- ▼ Resolver
 - VPCs

Hosted zones (1)

[View details](#) [Edit](#) [Delete](#) [Create hosted zone](#)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value

	Hosted zone name	Type	Created by
☐	asnae-ust-cloudops-domainas...	Public	Route 53

0 hosted zone selected

Select a hosted zone to see its details

Created Health checks

Route 53 > Health checks

Route 53

- Dashboard
- Hosted zones
- Health checks
- Profiles [New](#)
- ▼ IP-based routing
 - CIDR collections
- ▼ Traffic flow
 - Traffic policies
 - Policy records
- ▼ Domains
 - Registered domains
 - Requests
- ▼ Resolver

Health checks (2) Info

[Create health check](#)

Route 53 health checks monitor the health and performance of your application's servers and endpoints.

Find health check

	ID	Name	Details	Status in last 24 hours	Current	Actions
☐	b54b7e01...	secondary...	http://clo...		Health	
☐	cb1ed290...	primary-a...	http://a6...		Health	

Route 53

Dashboard

Housed zones

Health checks

Profiles New

▼ IP-based routing

CIDR collections

▼ Traffic flow

Traffic policies

Policy records

▼ Domains

Registered domains

Requests

▼ Resolver

Records (4)

DNSSEC signing

Hosted zone tags (0)

Records (4) Info

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value

Type

Routing p...

Alias

1

Record ...

Type

Routin...

Differ...

Alias

Value/Route traffic to

TTL (s...

☐	asnae-ust...	NS	Simple	-	No	ns-1932.awsdns-49.co.uk. ns-1140.awsdns-14.org. ns-524.awsdns-01.net. ns-313.awsdns-39.com.	172800
☐	asnae-ust...	SOA	Simple	-	No	ns-1932.awsdns-49.co.uk. a...	900
☐	project.as...	A	Failover	Primary	Yes	a6bb5671a27984aa3901f09...	-
☐	project.as...	A	Failover	Secondary	Yes	cloudops-demo-alb-188383...	-

Brought a domain from go daddy which is “project.asnae-ust-cloudops-domainasna.xyz”

Route 53 hosted zone, create an Alias record (A type) for application

Alias record should point directly to the DNS name of the AWS Application Load Balancer that is fronting Kubernetes Ingress.

Asnae UST Cloud Op...

Dashboard

Domain

Website

Email

Store

Appointments

Social

Marketing

Conversations 1

Customers

Deals

Domain

asnae-ust-cloudops-domainasna.xyz

Connect Domain

Overview

DNS

Registration Settings

Products

DNS Records

Forwarding

Nameservers

Hostnames

DS Records

Nameservers determine where your DNS is hosted and where you add, edit or delete your DNS records.

Using custom nameservers

Change Nameservers

Nameservers

ns-1932.awsdns-49.co.uk

ns-1140.awsdns-14.org

ns-524.awsdns-01.net

We serve cookies. We use tools, such as cookies, to enable essential services and functionality on our site and to collect data on how visitors interact with our site, products and services. By clicking Accept, you agree to our use of these tools for advertising, analytics and support. [Privacy Policy](#)

Manage

Decline

Accept

X

Accessing the application using domain name

CloudOps Demo - Three-Tier Architecture

Frontend --> Backend --> Database

Add New User

Name

asna

Email

+ ADD USER

Users (3)

us-west

us-west@gmail.com

shameel

shameel@gmail.com

asna

Asna.E@ust.com

Architecture Overview

This application demonstrates a three-tier architecture running on AWS EKS:

Frontend Tier: React application with Material-UI (Port 3000)

Backend Tier: Node.js/Express REST API (Port 5000)

Database Tier: PostgreSQL database (Port 5432)

Phase5 :- Cloudwatch Monitoring

CloudWatch

Favorites and recents

AI Operations

Alarms

▼ Logs

Log groups

Log Anomalies

Live Tail

Logs Insights

Contributor Insights

▼ Metrics

Application Signals (APM)

Network Monitoring

Log groups (9)

By default, we only load up to 10000 log groups.

/aws/conta

4 matches

Exact match

Log group	Log class	Anomaly d...	Da...	Se...	Reten...
/aws/containerinsights/cloudops-demo/application	Standard	Configure	-	-	Never
/aws/containerinsights/cloudops-demo/dataplane	Standard	Configure	-	-	Never
/aws/containerinsights/cloudops-demo/host	Standard	Configure	-	-	Never
/aws/containerinsights/cloudops-demo/performance	Standard	Configure	-	-	Never

Container Insights

Service: EKS

Add to dashboard

View in maps

View performance dashboards

Clusters state summary (1)

As of June 24, 2025, 11:34 am (UTC+05:30)

Cluster

cloudops-demo

Utilization

Alarm states per resource type

Cluster

No alarms detected

Performance and status summary

Last 1 min

Clusters CPU (avg)

Utilization 0%

Reserved 0%

Clusters Memory (avg)

Utilization 0%

Reserved 0%

Pods (sum)

Desired 0

Ready 0

Nodes (sum)

Unavailable 0

Available 3

Control plane summary

Last 3 hours

Phase-6 – Enabled email notification when any stage in code pipeline failed

Amazon SNS

Dashboard

Topics

Subscriptions

▼ Mobile

Push notifications

Text messaging (SMS)

New Feature

Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

newtopic

Edit

De

Details

Name

newtopic

ARN

arn:aws:sns:us-east-1:715841362372:newtopic

Type

Standard

Display name

-

Topic owner

715841362372

Amazon EventBridge > Rules > newrule

Amazon EventBridge <

Dashboard [New](#)

▼ Developer resources

- Learn
- Sandbox
- Quick starts

▼ Buses

- Event buses
- Rules**
- Global endpoints
- Archives
- Replays

▼ Pipes

- Pipes

▼ Scheduler

Rule details [Info](#)

Rule name newrule	Status Enabled	Event bus name default	Type Standard
Description	Rule ARN arn:aws:events:us-east-1:715841362372:rule/newrule	Event bus ARN arn:aws:events:us-east-1:715841362372:event-bus/default	

Event pattern [Info](#) [Edit](#)

```
1 {
2   "source": ["aws.codepipeline"],
3   "detail-type": ["CodePipeline Stage Execution State Change"],
4   "detail": {
5     "state": ["FAILED", "CANCELED", "SUCCEEDED"]
6   }
7 }
```

Lambda > Functions > newfunction

Code source [Info](#) [Upload from](#)

EXPLORER

- NEWFUNCTION
 - lambda_function.py

DEPLOY

Deploy (Ctrl+Shift+U)

Test (Ctrl+Shift+I)

TEST EVENTS [NONE SELECTED]

+ Create new test event

newfunction

```
1 import json
2 import boto3
3
4 sns_client = boto3.client('sns')
5 codepipeline_client = boto3.client('codepipeline')
6
7 # Replace with your actual SNS Topic ARN
8 SNS_TOPIC_ARN = 'arn:aws:sns:us-east-1:715841362372:newtopic'
9
10 def lambda_handler(event, context):
11     print("Received event:", json.dumps(event))
12
13     detail = event.get('detail', {})
14     pipeline_name = detail.get('pipeline', 'unknown-pipeline')
15     execution_id = detail.get('execution-id', 'unknown-execution')
16     state = detail.get('state', 'UNKNOWN')
17     failed_stage = 'Not determined'
18
```




AWS Notifications <no-reply@sns.amazonaws.com>



To: Asna Eranthodi(UST,IN)

Tue 6/24/2025 7:07 AM

External Email

Do not click any links or open any attachments unless you trust the sender and know the content is safe.

Pipeline 'event-pipe' has FAILED.
Execution ID: 152d1680-09ef-4c2a-9704-313062336339
Failed Stage: Error fetching stage: 'stageStates'
State: FAILED

--

If you wish to stop receiving notifications from this topic, please click or visit the link below to unsubscribe:

[https://urldefense.com/v3/https://sns.us-east-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:us-east-1:715841362372:newtopic:efb7c806-c66f-4dd1-ab14-5fa1804a1d35&Endpoint=290419@ust.com;!!PdM5GIU!VafHogKT9QR-E5z8o49OJi0SUJo7tX5NFQVFaoJWbUY7u_Z5_w7H3tjBZO5PgW31laHri8Bxrjcnh3RptPUjQb8\\$](https://urldefense.com/v3/https://sns.us-east-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:us-east-1:715841362372:newtopic:efb7c806-c66f-4dd1-ab14-5fa1804a1d35&Endpoint=290419@ust.com;!!PdM5GIU!VafHogKT9QR-E5z8o49OJi0SUJo7tX5NFQVFaoJWbUY7u_Z5_w7H3tjBZO5PgW31laHri8Bxrjcnh3RptPUjQb8$)

Please do not reply directly to this email. If you have any questions